SemEval-2026 Task 7: Everyday Knowledge Across Diverse Languages and Cultures

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Abstract

Large language models (LLMs) often struggle with culturally grounded knowledge of everyday life, particularly for regions and languages that are underrepresented in large-scale training data. Existing LLMs are trained on information from the internet, especially from Wikipedia, which may have inaccuracies or a western bias when discussing aspects of different cultures. While modern LLMs perform well on general question answering, they frequently lack access to common cultural knowledge that is rarely documented in widely available online resources. We propose a retrieval-augmented generation (RAG) approach for culturally aware short answer question answering. Our system is evaluated on Track 1 of SemEval-2026 Task 7 using the BLEnD benchmark. Experimental results demonstrate that retrieval-augmented prompting improves the generation of concise, culturally grounded answers across multiple regions. We report accuracy, precision, recall, and F1 scores for selected countries.

1 Introduction

Question answering is often regarded as a largely solved problem, as modern large language models can typically produce adequate responses to a wide range of factual and commonsense queries. Simple prompting strategies are frequently sufficient to elicit correct answers for questions grounded in well-represented domains and languages. However, this apparent success masks substantial limitations when questions involve culturally specific knowledge or regions that receive comparatively less coverage in large-scale training data.

Cultural question answering becomes particularly challenging for languages, dialects, and communities that are underrepresented relative to dominant “prestige” varieties. For example, in the case of Arabic, Egyptian Arabic has achieved cultural prominence through media and economic influence, often at the expense of regional varieties that receive significantly less representation. As a result, culturally grounded knowledge associated with smaller dialects or localized practices may be absent or only weakly encoded in pretrained models.

While it is infeasible to directly retrain or substantially modify large language models to address these gaps, we explore the use of Retrieval-Augmented Generation (RAG) as a practical alternative. By supplying models with external, region-specific contextual information at inference time, we aim to “nudge” latent representations toward culturally relevant knowledge that may otherwise remain inaccessible. This additional context can help surface culturally appropriate answers even when such information is not explicitly present in the model’s default outputs.

More concretely, we hypothesize that providing background information framed from the perspective of individuals within a given region can guide models toward outputs that better reflect local customs and everyday knowledge. Through retrieval-augmented prompting, our system seeks to bridge the gap between globally trained language models and culturally specific question answering scenarios.

2 Task Description

SemEval-2026 Task 7 Track 1 focuses on Short Answer Question (SAQ) generation for culturally grounded everyday knowledge. Given a natural language question and a target country or region, systems must generate a brief, culturally appropriate answer reflecting common local practices.

System outputs are evaluated against multiple human-annotated reference answers using exact match–based metrics, including precision, recall, and F1, as defined by the BLEnD benchmark.

3 Related Work

3.1 Prompting-Based Question Answering with LLMs

Recent progress in question answering has been driven largely by the emergence of large language models and prompting-based approaches. Instruction-tuned models have demonstrated strong zero-shot and few-shot performance across a wide range of QA tasks, often eliminating the need for task-specific training data. While these models are effective for general knowledge questions, prior work has shown that they tend to favor training data from high-resource languages and dominant cultures represented on the internet. As a result, responses to culturally specific or regionally grounded questions—particularly those involving smaller dialects—are often incomplete, overly generalized, or biased toward Western perspectives.

3.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has been proposed as a means of addressing some of these limitations by incorporating external documents into the generation process. By retrieving relevant textual evidence and conditioning generation on this information, RAG systems have been shown to improve factual accuracy and reduce hallucinations in knowledge-intensive tasks such as open-domain question answering. Prior work has primarily focused on factual correctness and coverage, with comparatively little research examining the use of retrieval to enhance cultural specificity or surface localized linguistic patterns, especially for underrepresented dialects. This gap motivates our exploration of RAG as a tool for culturally grounded question answering rather than purely factual augmentation.

3.3 Cultural and Multilingual Evaluation Benchmarks

Evaluation in cultural and multilingual settings introduces additional challenges. While several benchmarks exist for cross-lingual and multilingual evaluation, they often rely on standardized language forms that do not fully capture the nuances of regional dialects or culturally grounded knowledge. Consequently, purely quantitative metrics may fail to reflect cultural appropriateness or contextual accuracy. Prior work has therefore emphasized the importance of qualitative analysis when assessing cultural alignment, particularly in low-resource settings. In this work, we aim to complement automatic evaluation metrics by closely examining the retrieved content and the cultural relevance of generated answers.

4 System Overview

4.1 Retrieval-Augmented Generation

We built in RAGs to pull information from Wikipedia and Britannica to hopefully provide more context to the large language model as they are relatively high quality and diverse in their information offerings from within a vector database based on ChromaDB which by default utilizes the all-MiniLM-L6-v2 model running on Onnx for embeddings. A small limitation is that the RAG introduces a significant amount of latency as the library from Wikipedia is unofficial and hasn't been maintained for a long time, which as a result is quite slow and outdated. Then there are the speed issues with retrieving information from the vector database which runs the database locally. Repeated queries are sped up however as we cache the queries that are made so that we do not attempt to scrape multiple times for the same article. Queries are processed by another generation by a LLM, currently Llama 4 Maverick to convert the question into a list of possible page titles to explore. (Encyclopedia Britannica, 2025e,f,a,d,b,c).

4.2 Prompting Strategy

Using prompt engineering, we used a method of roleplaying as a guide and tourist interaction so that there would be an incentive to share basic information while utilizing limited vocabulary to fit the typical outputs of the

BLeND dataset. The ideology behind building a guide and tourist interaction is that a guide would be incentivized to share their local culture without overburdening like how a researcher much do with context. It also removes overly specific categories and more of a personal tone that a regular person would be looking for. We also created a slight "language learning" tone so that the LLM would provide not only the answer in the foreign language but also choose words of simplicity and generalizable across the population instead of an overly specific word like how one can say the "Spotify Camp Nou" but that would be likely be open to interpretation for various people but instead "estadio" might fit the criterion better.

4.3 Model Backend

We first attempted various open source models and ultimately settled on Llama 4 Maverick due to its multilingual training data, low cost of usage and open source architecture, handled through OpenRouter for the simplicity of testing out various models. We did start with attempting Transformers but quickly realized that we were running out of local GPU resources, then attempted to use Gemini but then ran into Free Tier utilization limits. This provides a more modern interpretation with newer models compared to the LLama 3 70B model that was tested as a part of the original paper.

4.4 Answer Selection and Postprocessing

We did not perform much post-processing on the outputs besides ensuring the output form factor matched what was seen in the BLEnD benchmark such as removing trailing periods, ensuring that all outputs were in lower case, removing quotation marks which would show up from time to time. A lot of the issues were solved by providing what we were looking for in the prompt.

5 Experimental Setup

5.1 Datasets

The original data was provided by BLEnD. The BLEnD data was a handed-crafted benchmark dataset designed to evaluate LLMs' knowledge of common-sense across different cultures and languages. The data was collected across 16 countries/regions from native speakers among different aspects such as food, education, education, et al. They collected those data by questioning interviewees and removed terms and stereotypes for any single countries. Each answers were annotated by at least 5 native speakers who have lived long-term within the target culture. Multiple answers and "I don't know" were allowed as well. They also have provided all questions, prompts, and answers translated into English. And we selected Mexico, Spain, South Korea, Nigeria, China for our evaluations for experiment.

5.2 Evaluation Metrics

Since the task allowed multiple answers, we directly compare the responses generated by the LLMs with those in the dataset. Once the generated answer matchs one ini the dataset, it is considered correct. Based on this standard, we can also calculate overall F1, Recall, and Precision. F1 Score formula: F1 = 2 * (Precision * Recall) / (Precision + Recall) Precision: TP / (TP + FP) = correct predictions / total predictions Recall: TP / (TP + FN) = correct predictions / total ground truth. For experiments, we select questions by the country name and only use the translated (English) questions and prompts. Then we compare the model generated answers and the ground truth answers from corresponding country, then we are able to calculate the overall precision, recall, and F1.

6 Results

We tested our model in three different scenarios to evaluate the effectiveness of our RAG based approach. The first configuration served as a baseline, using solely the role playing prompt without any information retrieval. The second introduced RAG using snippets from relevant Wikipedia pages, and the third was a combination of the second approach but in addition, providing context using a snippet pulled from hand picked data about these countries we added into a txt file. Surprisingly, the Wikipedia-only configuration underperformed relative to the baseline, observing an accuracy decrease from 11% to 9%. This

Test	F1	Precision	Recall	Accuracy	Match
Simple prompting (LLaMA4)	0.06169	0.1340	0.04272	0.1340	67/500
Simple Wiki RAG	0.0560	0.1180	0.0394	0.1180	59/500
Simple Wiki RAG (trial 2)	0.0442	0.0920	0.0324	0.0920	46/500
Wiki RAG + hand picked	0.0753	0.1584	0.0535	0.1584	73/461

Table 1: Performance with varying types of RAG

suggests that the general data may introduce some noise or dilute the culturally relevant signals that the model already had. However, the hybrid configuration yielded our best performance, increasing accuracy up to around 16%. This improvement may show that the general knowledge bases may not be effective on their own, but when combined with high quality hand picked cultural data, they enhance the model's ability to interpret culturally grounded queries significantly. The F1 Score, Precision, Recall and Accuracy all tell a similar story, which is that the hybrid configuration worked notably better than the baseline model results. These different RAG configurations were only tested on the Spanish portion of the BLeND dataset to keep it consistent. The final run only completed 461 out of 500 questions due to time constraint but results are still comparable with the different metrics.

7 Conclusion and Discussion

7.1 Limitations and what went well

A major limitation of our approach stemmed from the datasets provided by the organizers. We were given only validation and test sets, with no accompanying training data. As a result, we had to independently source documents to train our model. This process proved challenging, as it was difficult to find materials that were both culturally authentic and written by local authors. Consequently, much of our supplemental data was drawn from Britannica and Wikipedia, which, while credible and well-researched, may not fully reflect the linguistic and cultural nuances present in locally produced texts. The bright side is that with additional time, we could also utilize the other articles in the various languages of Wikipedia to bolster our relevant information pool but are currently utilizing a setup where the primary backend language is English due the simplicity of our own understanding for the debugging process.

We were restrained from reaching our full potential due to the inadequacies of the evaluation script alongside having to provide our own training data given that the SEMEVAL organizers forbade utilizing the validation set. There is also confusion over the evaluation taggings e.g. what does the counts mean for each of the approved outputs. The scripts also had issues in the original repository. This was somewhat resolved later on when organizers told us later on not to use the original repository but use another repository which was the updated version without being alerted. This caused some issues as we were simply not aware of the change happening and was side-tracked for a significant amount of time with the evaluation script not working. However, even with the fixes, it came too late to be worked well into our current infrastructure.

7.2 Improvements and Future Work

Future work could significantly benefit from expanding and diversifying the sources we use for training data. One direction we could take is by incorporating books written in the target language by authors from those countries. They would be able to provide richer linguistic patterns, culturally grounded narratives, and authentic perspectives that are often missing from generalized online sources. Regardless of the text being fiction, non-fiction or children's literature, they tend to reflect cultural idioms, values and experiences. By integrating such materials, the model could gain a deeper understanding of cultural context and produce outputs that are more aligned with local norms and expressions.

We would like to test on many more languages, but given the time constraints and the high cost of prompting language models over 500 times per evaluation, we did our primary investigation in just Spanish. We could additionally try simply the hand picked RAG without the Wikipedia data, and see if there would be a noticeable difference in performance.

In addition to books, incorporating non-academic materials such as tourist brochures, cultural pamphlets, and locally distributed parenting guides would further enhance the dataset's authenticity and breadth. Tourist flyers often highlight cultural landmarks, traditions, and everyday practices, offering concise yet relevant information that can improve the model's background knowledge though must also be taken with a grain of salt given how "stereotypical" they can be alongside the agenda of invigorating businesses by exaggerating certain points to direct business. Since the Validation script has questions about the youth, parenting guides written by local authors can provide more depth, capturing family-oriented cultural expectations, social norms, and linguistic styles that are uniquely tied to the community.

8 Contributions

Isaac To programmed the LLM API framework using LangChain backend, provided OpenRouter API Key, created Question prompt, programmed vector database framework using ChromaDB backend, programmed Wikipedia RAG and search logic, programmed post processing, wrote abstract, wrote architecture section (RAG, prompting strategy, model backend, answer selection/post processing), wrote limitations.

Devin Shende defined object oriented code structure for main, model and rag files. Put the pieces all together: implemented main loop using Isaac's model implementation to prompt gemini for every item in the test script and save RAG info generated from Navodit's documents and wikipedia articles, and run script that calls Lucas' evaluation script to get our results. Wrote results section.

Nathan Perez created the report outline and gathered data from team members to organize the paper for the coding implementation of Overleaf. Contributed to the Abstract and Introduction part of the paper. Implementing the citation pipeline using BibTeX, including the creation and normalization of a custom bibliography (custom.bib) and the integration of in-text citations. Experimented with pre-processing of data.

Jiayuan She created the evaluation script, evaluation and experiment setup part of the report.

Navodit Maheshwari helped find documents used in the RAG model for various different countries and worked on the Abstract, Related Work and Conclusion part of the paper. Also helped with preparing data from the documents to be able to be used with the model.

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